

Lesson 8: The Binomial Theorem — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Write down the binomial coefficients (Pascal's triangle) for $n = 4$.
2. Write down the coefficients for $n = 5$.
3. How many terms are there in the expansion of $(a + b)^7$?
4. Evaluate $C(5, 2)$.
5. Expand $(1 + x)^3$.

Section 2 — Core practice

6. Expand $(1 + 2x)^4$.
7. Expand $(2 + x)^3$.
8. Find the coefficient of x^2 in $(1 + 3x)^4$.
9. Find the coefficient of x^3 in $(1 + x)^6$.
10. Expand $(1 - x)^4$.
11. Find the coefficient of x in $(1 + x)^5$.

Section 3 — Challenge & exam-style

12. Find the term in x^2 in the expansion of $(3 + x)^5$.

13. Find the coefficient of x^4 in $(2 + x)^6$.

14. Find the constant term (term in x^0) in $(2 + x)^5$.

15. In $(1 + 2x)^5$, find the coefficient of x^3 .